

Computer Vision

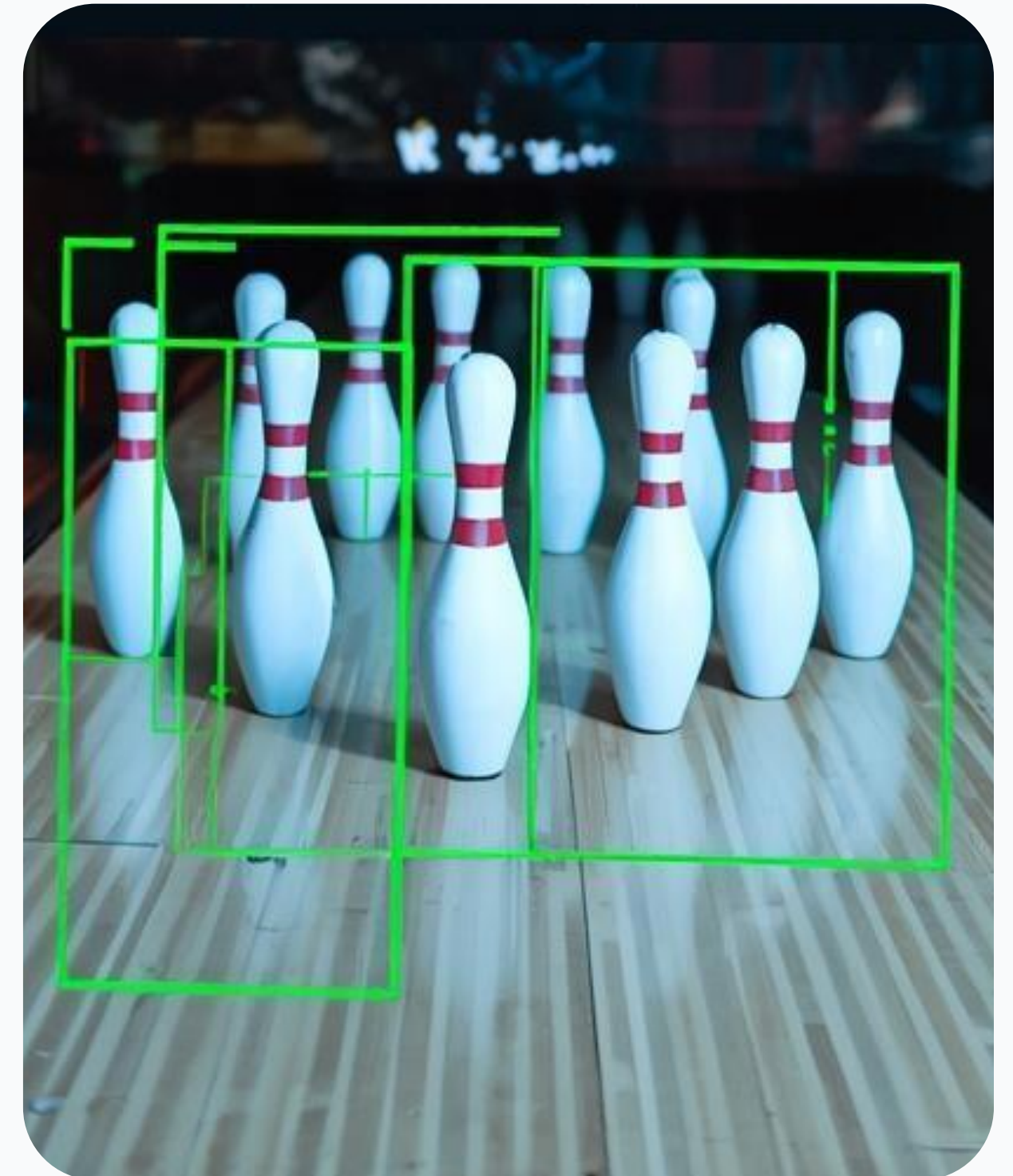
Final Project

Bowling Pin Detection



Introduction

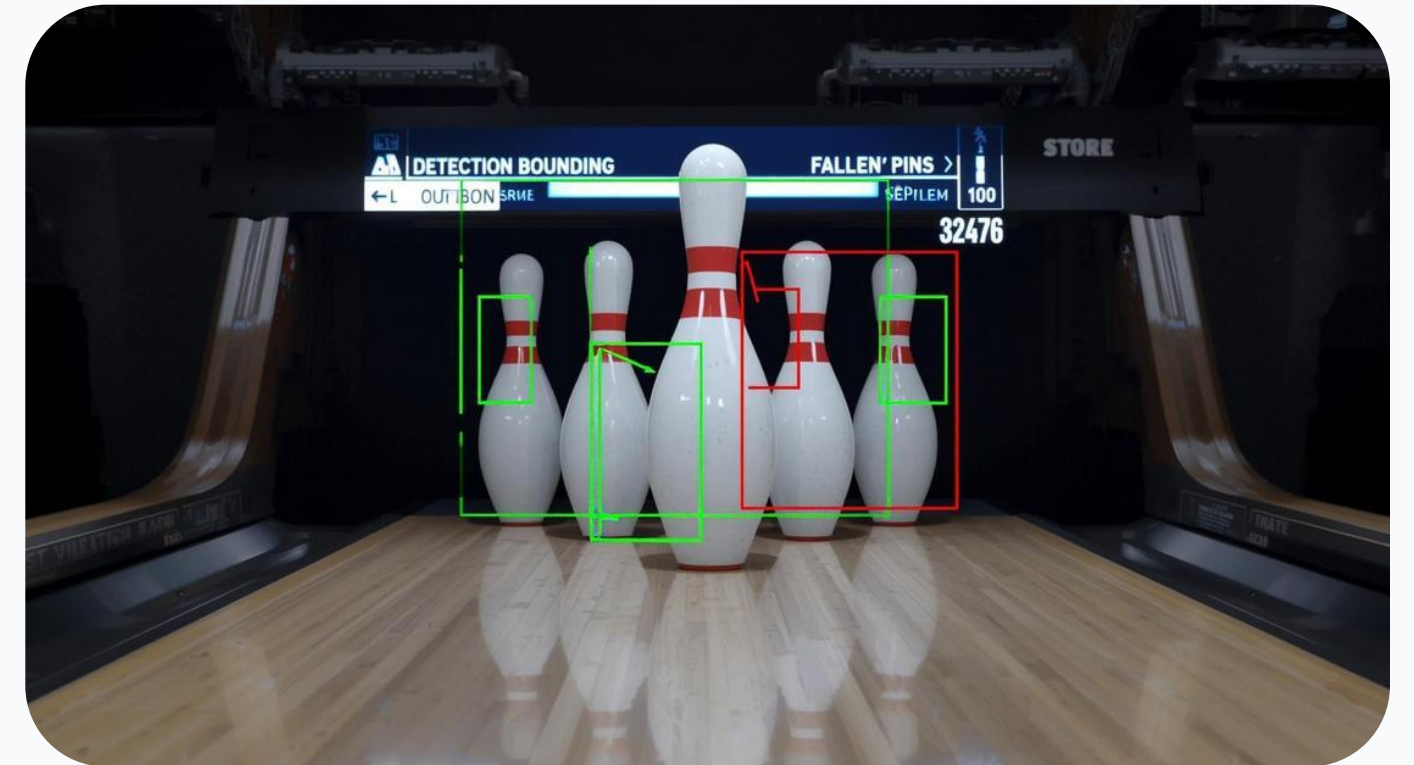
The project aims to develop automatic bowling score detection from mobile video using computer vision. The system combines a fine-tuned YOLOv8n detector with ByteTrack tracking and fall detection algorithms. Our dataset includes 412 annotated keyframes from a 30-second video capturing 7 throws.



Dataset and Annotation

The video frames were all extracted (900 frames), then the main ones, in which the ball hits the pins were selected (412 frame).

These keyframes were used to build the dataset then annotated using Roboflow. Two classes labeled: Standing_Pin and Fallen_Pin. Standard 10-pin rectangle layout used for consistent annotation.

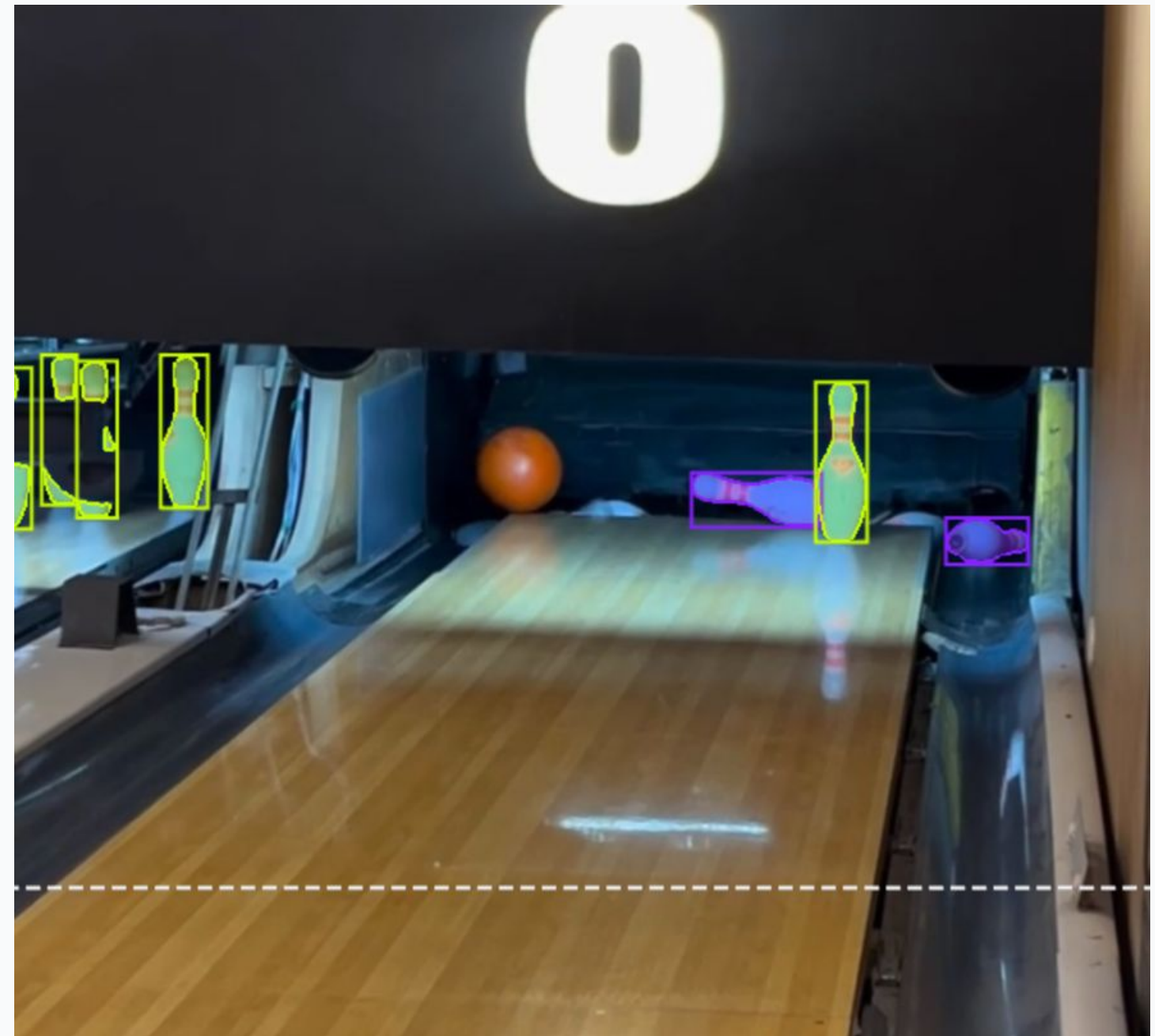


Video: 30 fps, 900 frames total

Annotations 412 keyframes with Standing_Pin & Fallen_Pin

Training Split: 80/20 split (330 train, 82 validation)

Dataset Annotation Sample

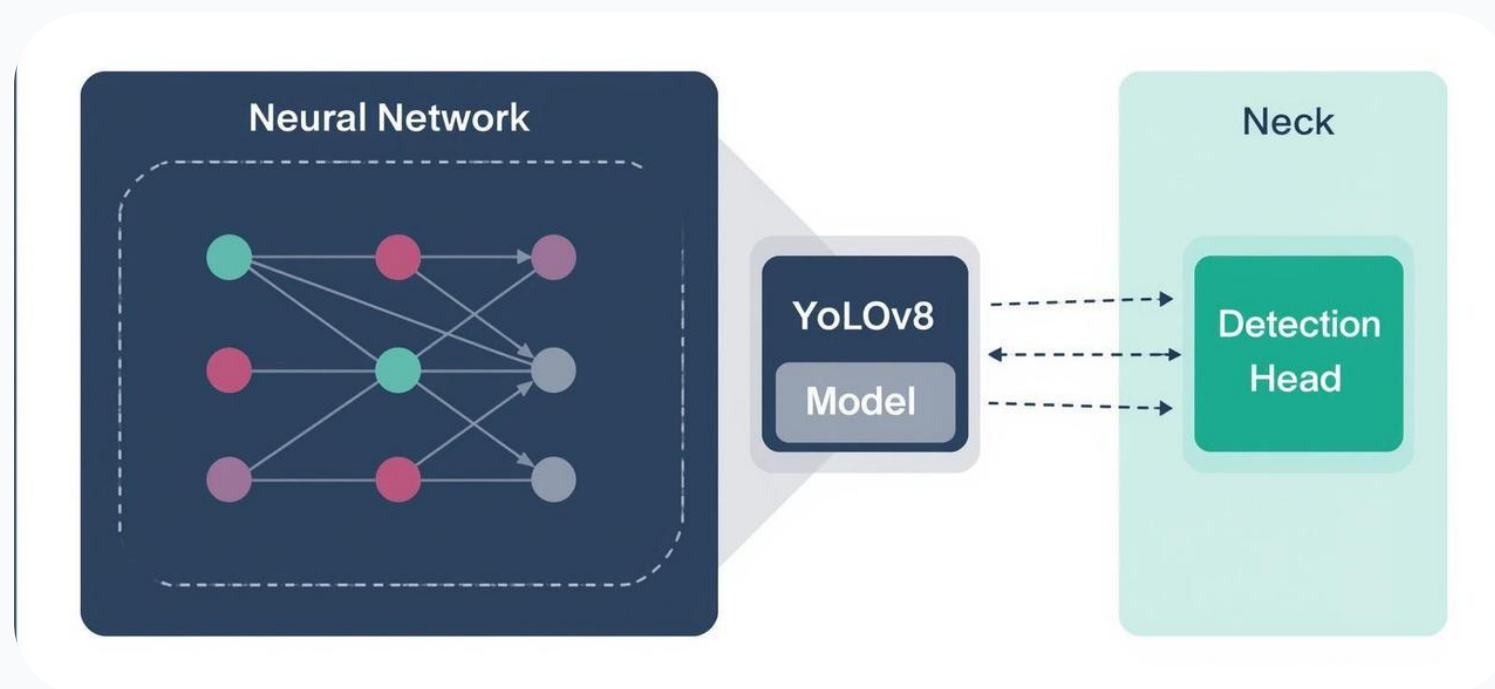


YOLOv8n Model Architecture

YOLOv8n was selected for its optimal balance of speed and accuracy. With only 3M parameters and 8.1 GFLOPs, it achieves ~8.1 ms inference per frame on Tesla T4. Pretrained on COCO and fine-tuned for bowling pin detection.

Key Features:

Two output classes: Standing_Pin, Fallen_Pin
Native ByteTrack integration for tracking
Lightweight with transfer learning capability



Training Setup and Hyperparameters

01

60 epochs with early stopping, image size 640×640, batch size 16

02

Optimizer: AdamW with learning rate 0.01

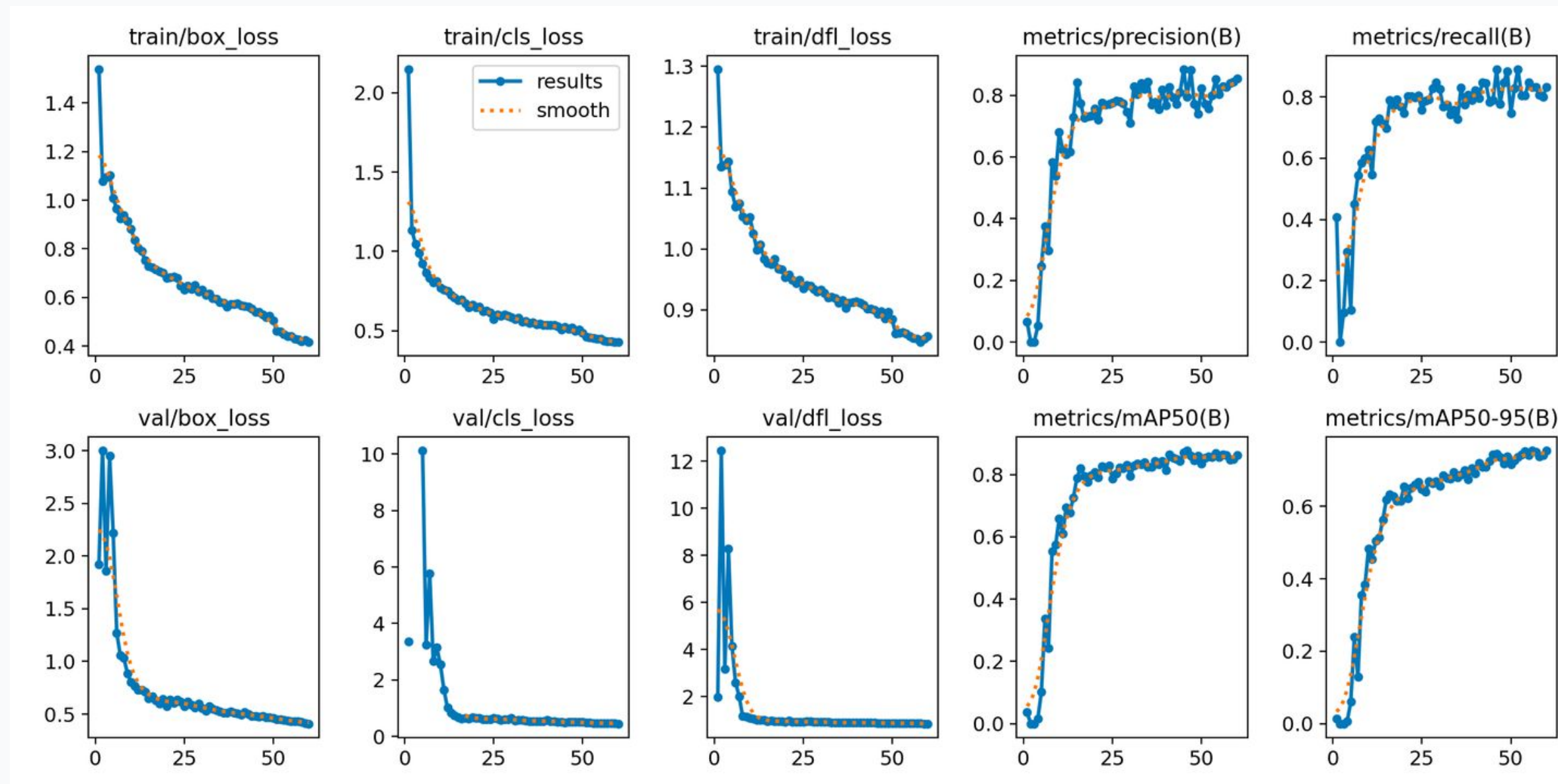
03

Data augmentation: horizontal flip enabled, vertical flip and rotation disabled

04

Smooth loss curves and no overfitting during training

Training Curves



Computer Vision Pipeline Overview



The 6-stage detection and scoring pipeline:

01

Frame ingestion using OpenCV at 30 fps + YOLOv8n detection with ByteTrack

02

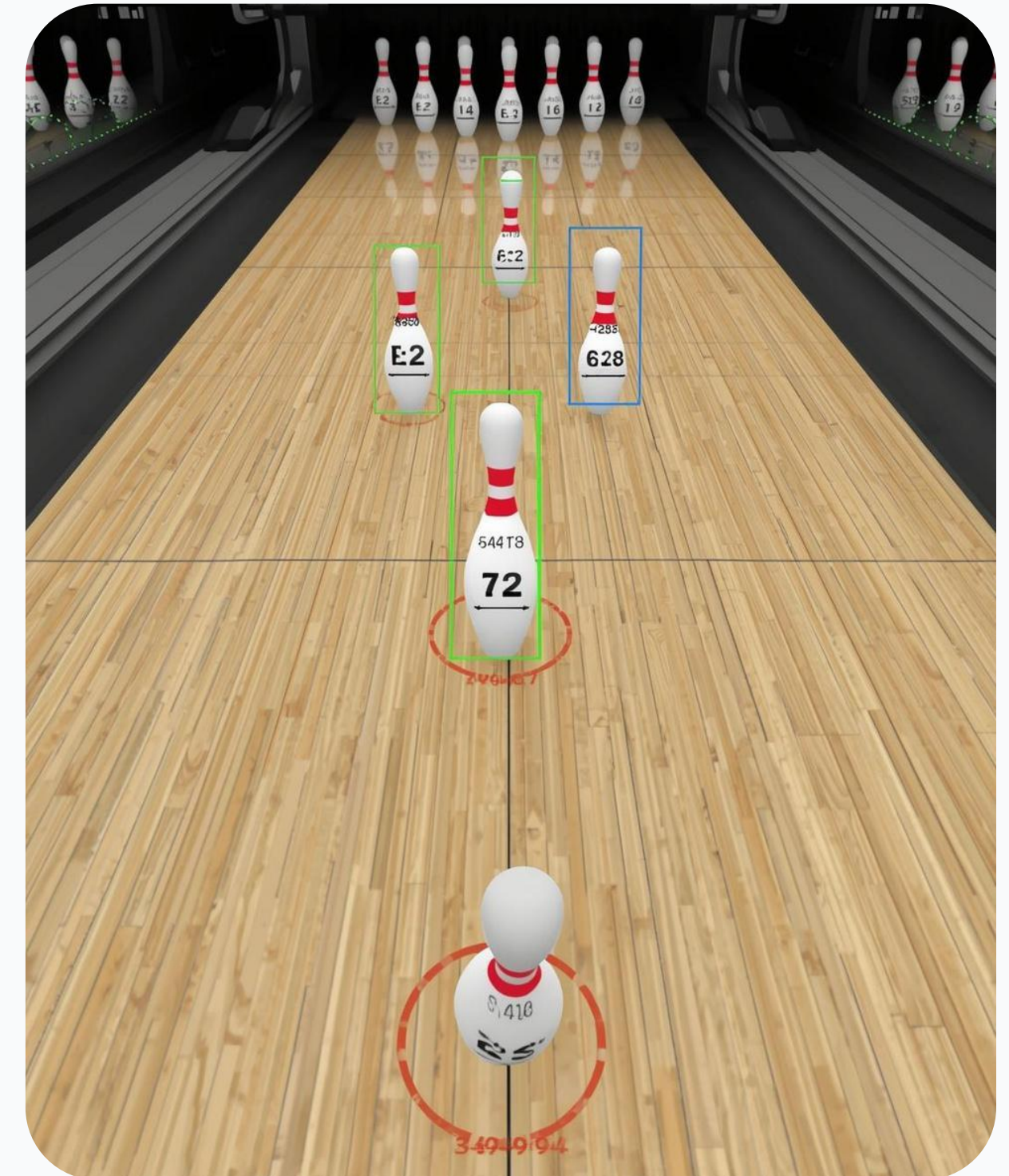
Fall detection: track ≥ 8 frames then absent ≥ 6 frames, warmup guard 1.0s

03

Annotation rendering: green boxes (standing), red circles (fallen) + summary frame

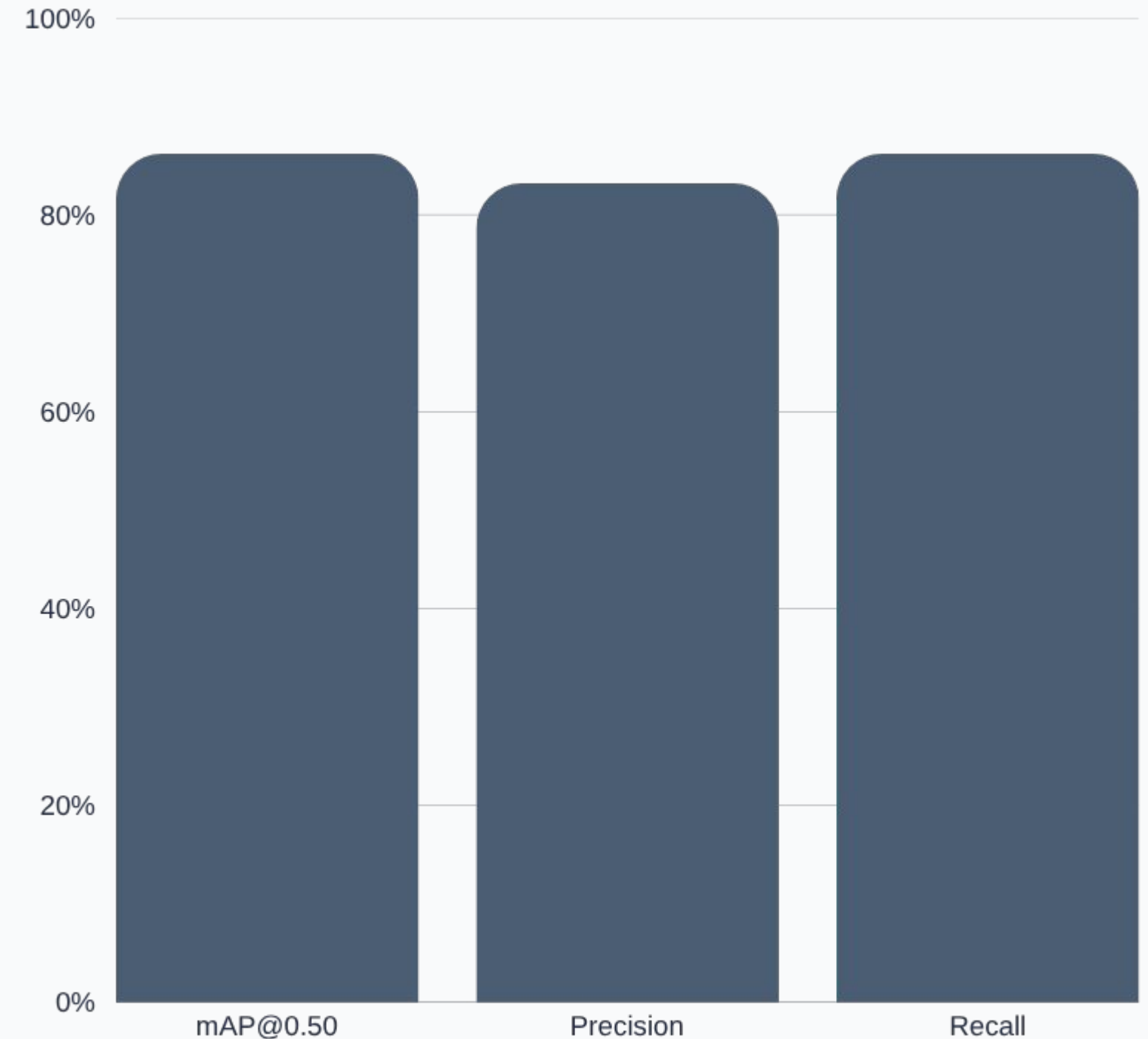
Tracking and Fall Detection

ByteTrack assigns unique IDs to each detected pin, with 108 unique track IDs generated across the test video. Fall detection triggers when a tracked pin is present for ≥ 8 frames then absent for ≥ 6 consecutive frames. Key challenges include ID resets between throws limiting cross-throw tracking continuity, and marker drift caused by occlusion and pin movement during gameplay.

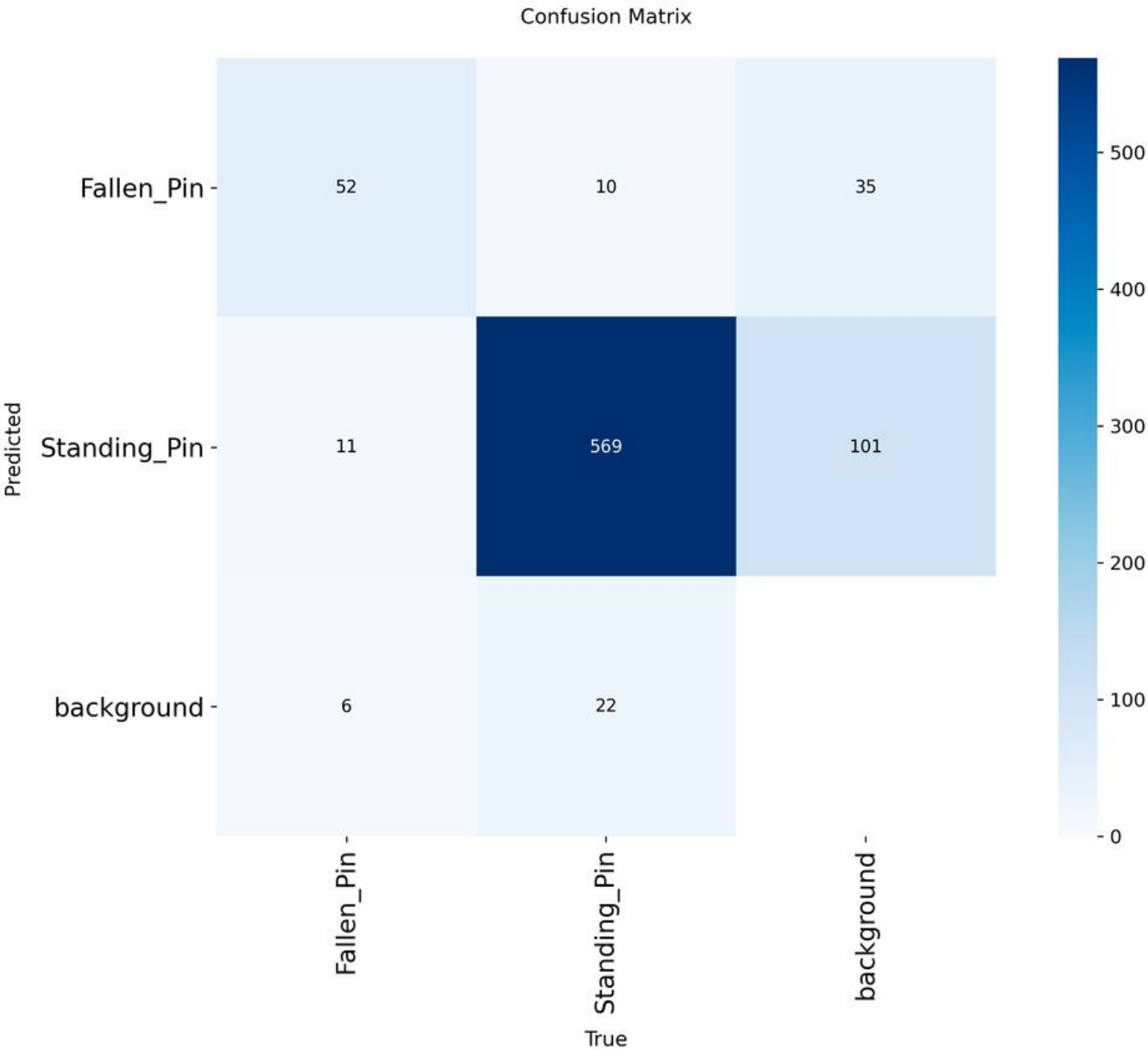


Model Performance Metrics

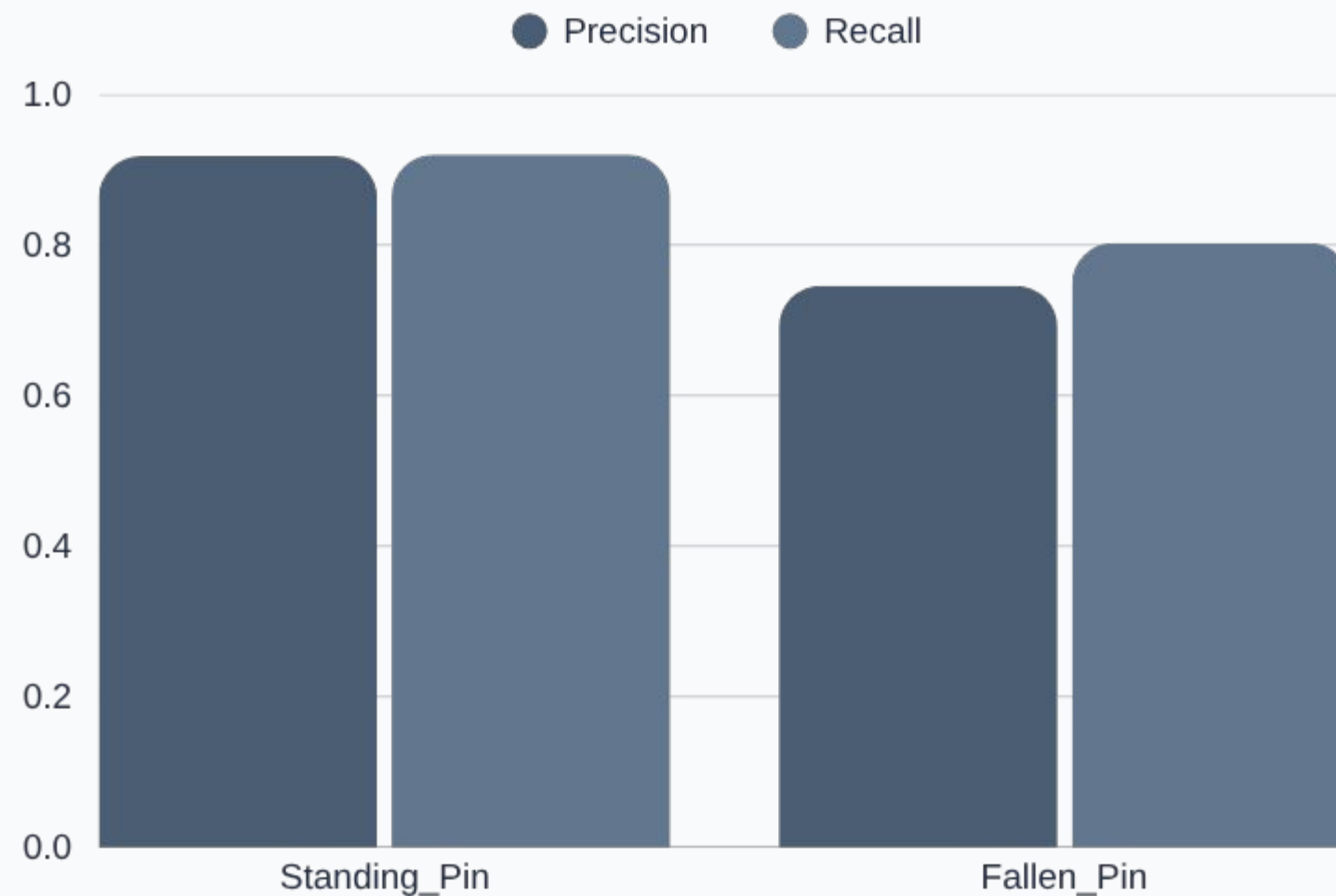
Our fine-tuned YOLOv8n model achieved excellent detection performance on the bowling pin dataset. The overall mAP@0.50 score of 0.862 demonstrates strong localization accuracy. With precision at 0.832 and recall at 0.862, the model effectively balances false positive reduction with comprehensive pin detection.



Confusion Matrix



Per-Class Metrics



Confusion matrix analysis reveals class-specific performance:

01

Standing_Pin: mAP 0.920, Precision 0.918, Recall 0.920

02

Fallen_Pin: mAP 0.803, Precision 0.746, Recall 0.803

03

Background false positives: 101 detections

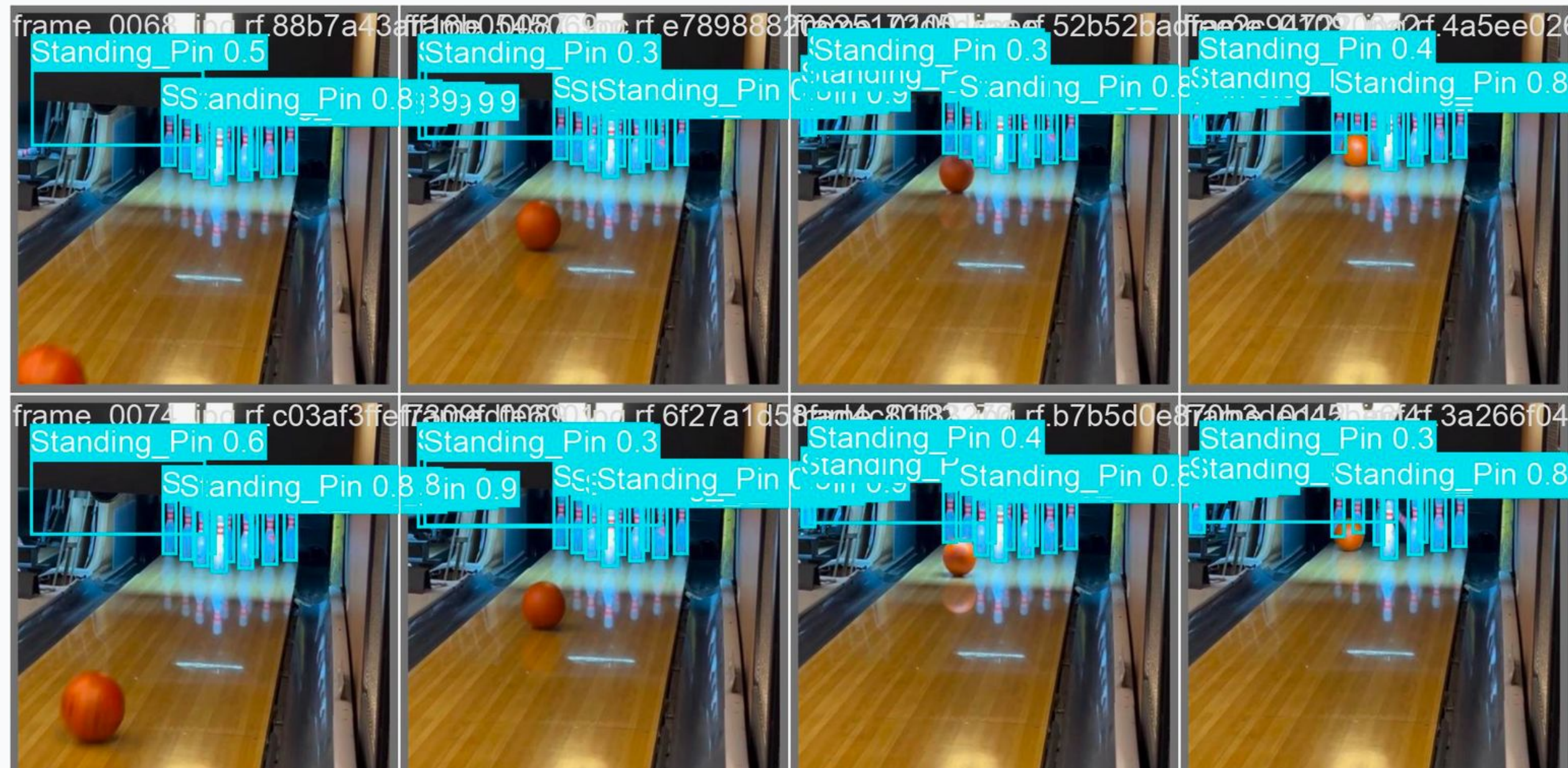
Standing_Pin: 94.7% correct (569/601) | Fallen_Pin: 75.4% correct (52/69)

Output and Scoring Visualization



The 30-second test video demonstrates real-time scoring across 7 bowling throws. The system identified 36 pin-fall events with precise timestamps, from the first fall at 1.1s to the last at 29.0s. Live score counter overlays update dynamically, while per-pin markers show detection status.

Sample Annotated Frames



Summary and Conclusions

Results:

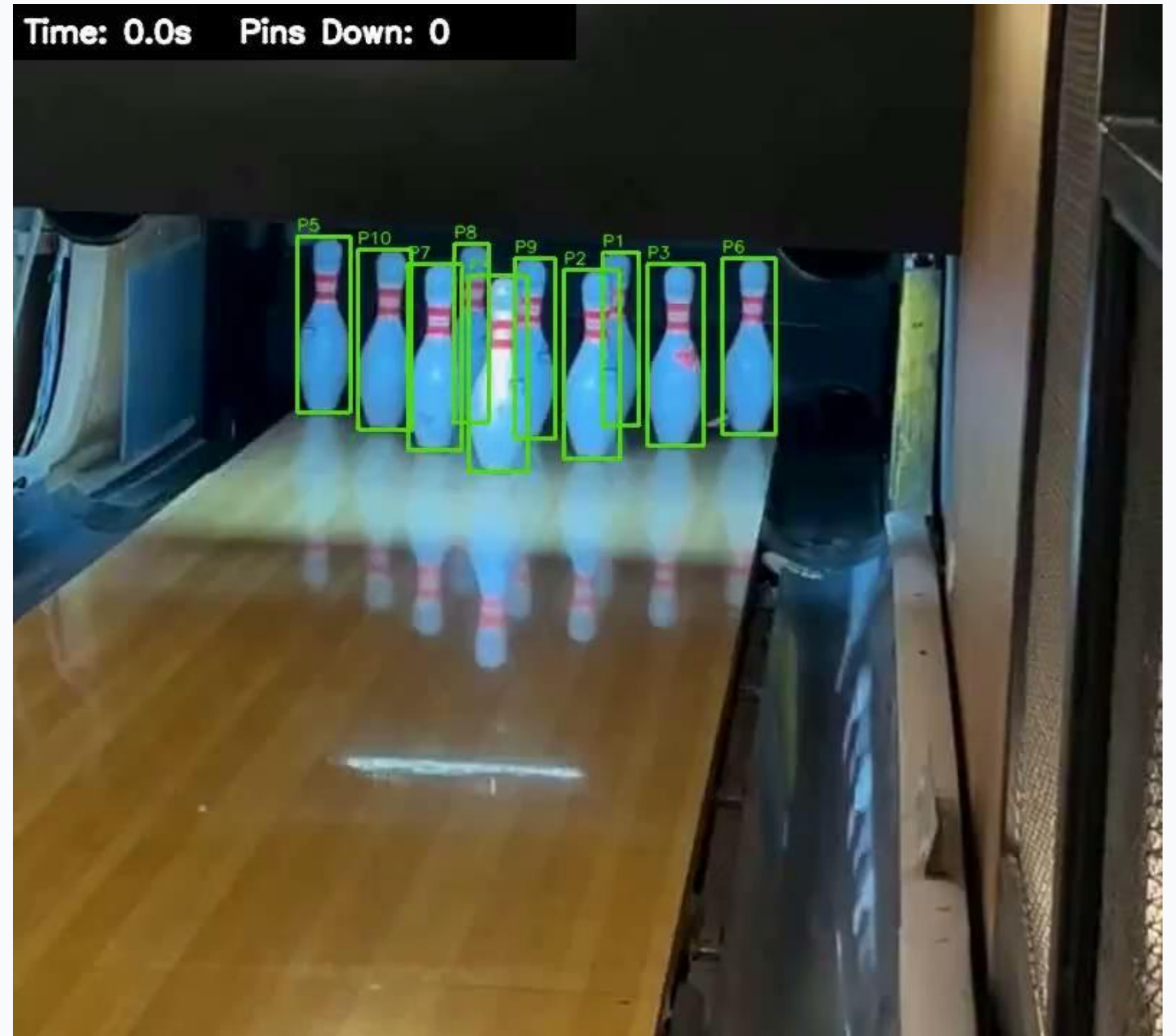
- 01** Developed end-to-end bowling pin detection and scoring pipeline

- 02** YOLOv8n fine-tuned with strong detection performance (mAP 0.862)

- 03** Effective tracking and fall detection with areas for improvement identified



Output Video



Thank You!

